Urban Growth Replicates, With Three Fixes

A replication of Duranton and Puga (2023)

Nora Fringe

Abstract

Duranton and Puga (2023, *Econometrica*) build a model in which incumbents use planning rules to cap city size, and find that relaxing permits in seven expensive U.S. cities would raise national output by 7.95%. We re-coded the authors' public replication files from scratch. Our code reproduces every headline number in the authors' output to six decimals. We confirm three coding errors. Only one moves a reported number much: the consumption loss from freezing cities at their 1950 sizes is 19.4%, not 26.8%. Los Angeles enters the seven-city policy by $22 under a price conversion that depends on an arbitrary regression choice, but every correct conversion keeps it in. Two further checks on the model's mechanism give mixed news. City edges spread with population at elasticity 0.41, not the model's 1. Urban costs still rise fast enough with population for cities to have a best size.

1 What we did

We rewrote the authors' public Stata programs in Python. Our regressions match the authors' log exactly: the block-group house value regression (128,620 block groups, root MSE 0.36674), the rent regression (133,249, 0.30794), and Table I columns (2) and (3) ($\gamma = 0.0769$ and 0.0726). The regressions using restricted NLSY79 and NHTS locations (Table II, Table I column 1) could not be re-run. An external audit prompted the check; every number below comes from our own code.

	Paper	Our code	Corrected
Output gain, seven cities expand (Table III)	7.95%	7.9463%	7.9463%
Consumption gain, same experiment	2.16%	2.1616%	2.1063%
GDP growth 1950–2010, actual / cities frozen	2.1% / 0.9%	2.086% / 0.921%	unchanged
Consumption loss after 60 years, cities frozen	26.8%	26.8153%	19.3901%
Growth from reallocation, fixed population	0.7 pp	0.658 pp	unchanged

Table 1: Headline numbers. "Our code" is our independent port run on the authors' public data. "Corrected" applies the fixes in Section 2. All nine cells of the paper's parameter-sensitivity table also reproduce.

2 Three coding errors

1. Converting log prices to dollars. The policy targets cities of over 3 million people whose periphery house price exceeds building cost by at least $200,000. The code converts predicted log prices to dollars by adding half the squared standard error of each city's dummy. The usual correction uses the variance of the regression residual. The dummy's standard error depends on which city is left out of the regression (Albany), so the dollar price changes when nothing economic does. Los Angeles clears the cutoff by $22. With Los Angeles as the omitted city, or with no correction, it misses by $302 and the output gain falls to 5.68%. Every standard correction (residual variance, Duan smearing, full prediction variance) puts Los Angeles between $226,000 and $229,000 and keeps the same seven cities (Figure 1, left). The headline survives; the selection rule is knife-edge.

2. The frozen-city consumption loss. In the model, newcomers to a city pay a permitting cost that leaves them with rural consumption. Section 7 of the code averages consumption that way. The frozen-city calculation instead credits all 2010 city residents, newcomers included, with incumbent consumption. Using the paper's own accounting, the loss is 19.4%, not 26.8%. The original figure is what one would get if permit payments were transfers handed back to everyone. Under that reading, though, the seven-city consumption gain would be 4.15%, not 2.16%. The paper's two numbers cannot both hold.

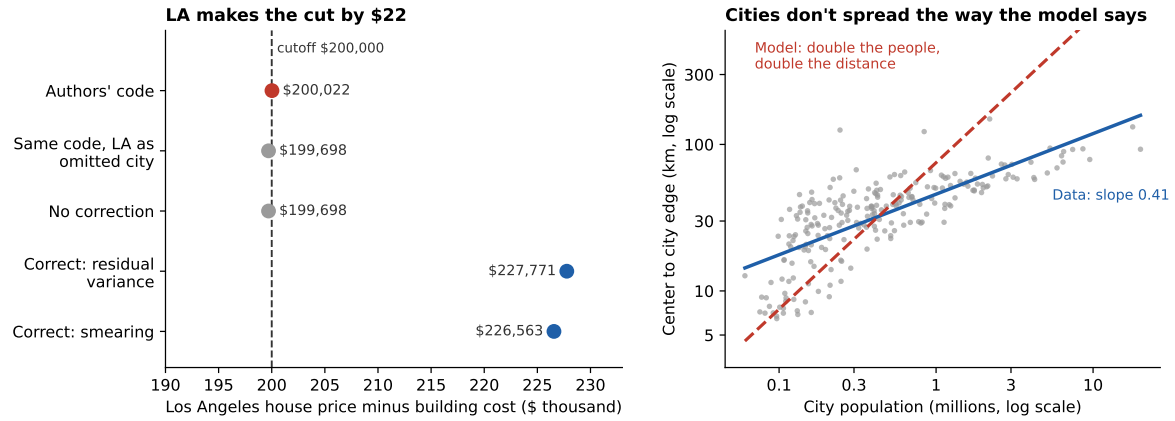


Figure 1: Left: Los Angeles' periphery house price minus building cost, under the authors' code and four alternatives. Gray: invalid conversions; blue: standard correct ones. Right: distance from center to city edge against population for 229 metro areas. The model's linear city requires the red slope of 1.

3. Closed cities vanish from the average. When the seven cities expand, ten small cities empty. Stata's `min()` ignores missing values, so their 774,360 former residents keep their incumbent count, are subtracted from the rural group, and appear in no average. Counting them as newcomers moves the consumption gain from 2.16% to 2.11%.

3 The theory, and two new tests

Equation (29) states that city sizes are exactly Pareto with shape $1/(1-\eta)$, where η is the reflecting floor. In discrete time this is generally false. With the floor at 0.8 and exponential growth shocks, the stationary tail exponent is exactly 2, not 5 (checked analytically and by simulation). Exact Pareto requires continuous time, as in Gabaix (1999). The paper's broader claim, that sizes approximately follow Zipf's law, survives. Smaller slips: the "homevoter" sentence after equation (24) holds only at the optimum, and equations (25) and (41) print h_{it} for h_t.

The model's mechanism needs urban costs to rise with population faster than agglomeration benefits: $\gamma + \theta = 0.11$ against $\sigma + \beta = 0.08$. It gets there through a linear city whose edge distance is proportional to population. In the authors' data, edge distance rises with population at elasticity 0.41 (s.e. 0.02), close to the 0.5 of a round city (Figure 1, right). Taken literally, this cuts the cost elasticity to about 0.07, below the agglomeration elasticity, and cities would have no best size. But the object incumbents care about can be measured directly: center house prices plus marginal consumption rise with population at elasticity 0.104 (s.e. 0.010), close to the model's 0.11. So the mechanism holds in reduced form, through channels other than the linear city. One estimator of γ (Table I, column 2) treats the highest-rent block as the center. That block sits a median 7 km out, where the model's rent schedule would imply a slope several times larger than the 0.077 found. That column is a reduced-form gradient, not a structural estimate.

4 Why it matters

Planning restrictions in America's most productive cities are a live policy question, and this paper's 8% output gain is a model-based estimate of what is at stake. That number replicates exactly from public data and survives every fix here. Readers should know three things. The consumption gain is about a quarter of the output gain, and freezing cities at their 1950 sizes would have cost a fifth of consumption, not a quarter. The seven-city policy hangs on a $22 margin for its largest city. And the model's geography does not match how cities actually spread, even though its key cost elasticity does.

Notes. The 0.104 elasticity is a cross-sectional OLS estimate and could be biased upward if amenities raise both population and prices. We have not checked for published comments on these points.

References. Duranton, G. and D. Puga (2023), Urban growth and its aggregate implications, *Econometrica* 91(6), 2219–2259, and replication files (diegopuga.org/data/urbangrowth). Gabaix, X. (1999), Zipf's law for cities: an explanation, *Quarterly Journal of Economics* 114(3), 739–767.